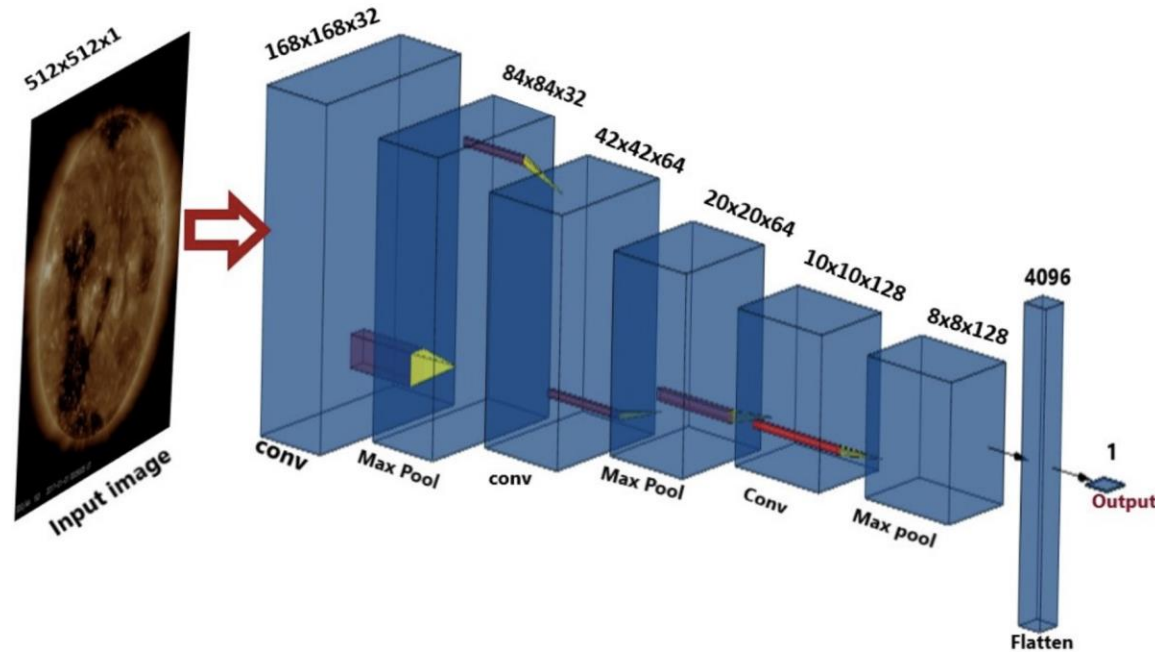


Raju, H., Das, S. CNN-Based Deep Learning Model for Solar Wind Forecasting. Sol Phys 296, 134 (2021)



Raju: Single SDO 193A image passes through CNN to predict 2 hr average SWS at L1
SDO images from SDO MLv1 dataset with resolution 512*512, SWS from omniweb by
computing the average of 12 mins averages from 2 hrs.

Training/validation are splitted once (no cv) for 2013~15 and 2017, and tested on 2018
Mapping of images and SWS during the training phase is be time delaying 1AU/observed
SWS, and during testing the model predict SWS 4 days after.

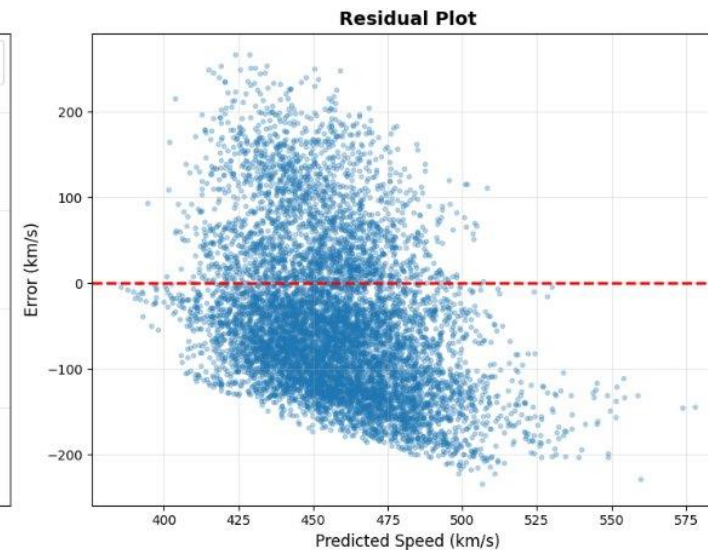
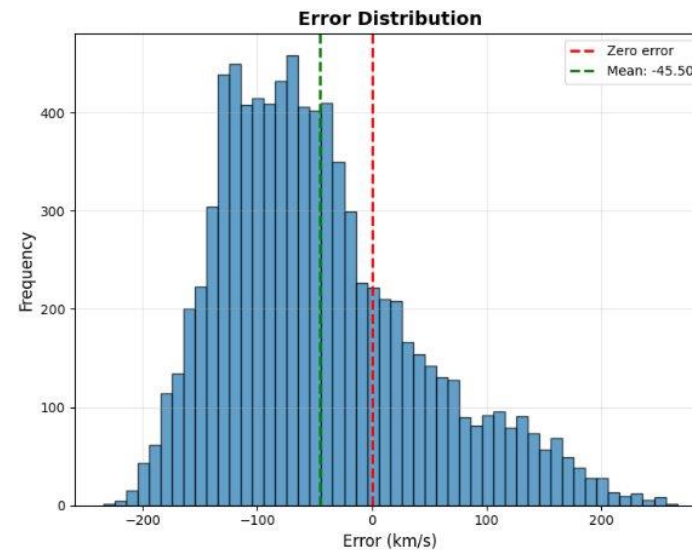
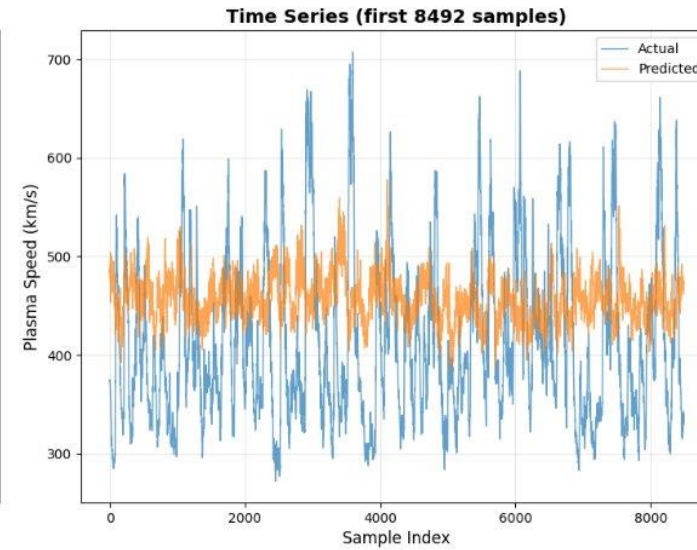
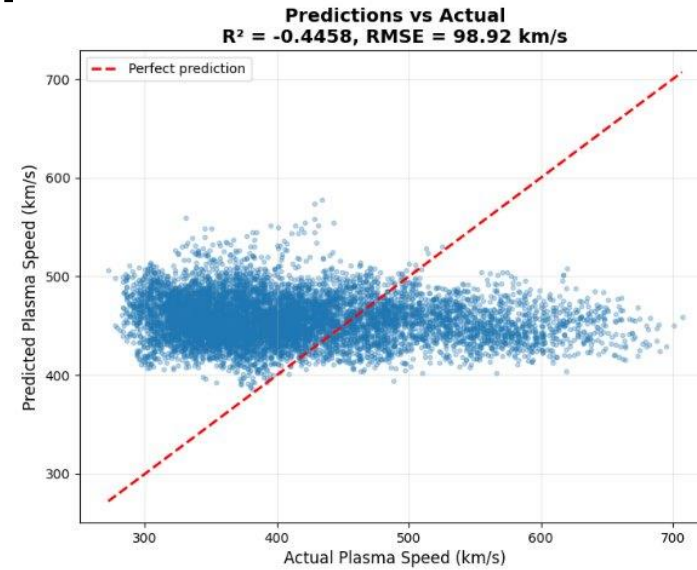
My implementation: SDO images from SDOMLv2 dataset with resolution 512*512, SWS 1hr average directly from omniweb.

Training/validation are splitted once (no cv) for 2013~15 and 2017, and tested on 2018

Mapping of images and SWS during the training phase is be time delaying 1AU/observed SWS, and during testing the model predict SWS 4 days after, and also 1AU/predicted SWS.

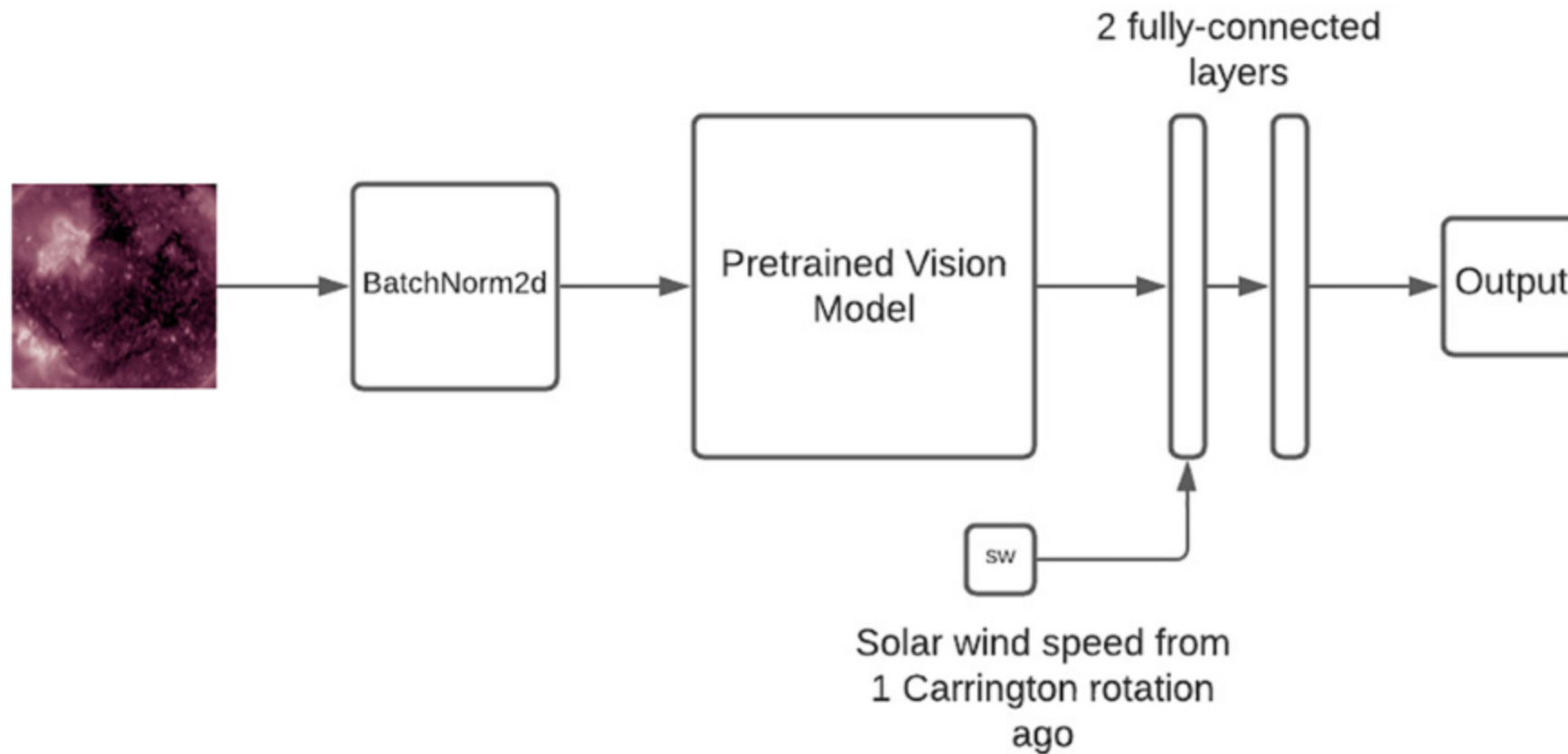
Method	RMSE km/s
Raju validation with dynamic lead	89~82
Raju test trained with dynamic lead, test with 4 days lead	76.3
Raju test trained with fixed 4 days lead, test with 4 days lead	78
My validation with dynamic lead	88.3/72.7
My test trained with dynamic lead, test with 4 days lead	98.9/99.7
My test trained with dynamic lead,, test with dynamic lead	110/110
1 day persistence	77
27 days persistence	94

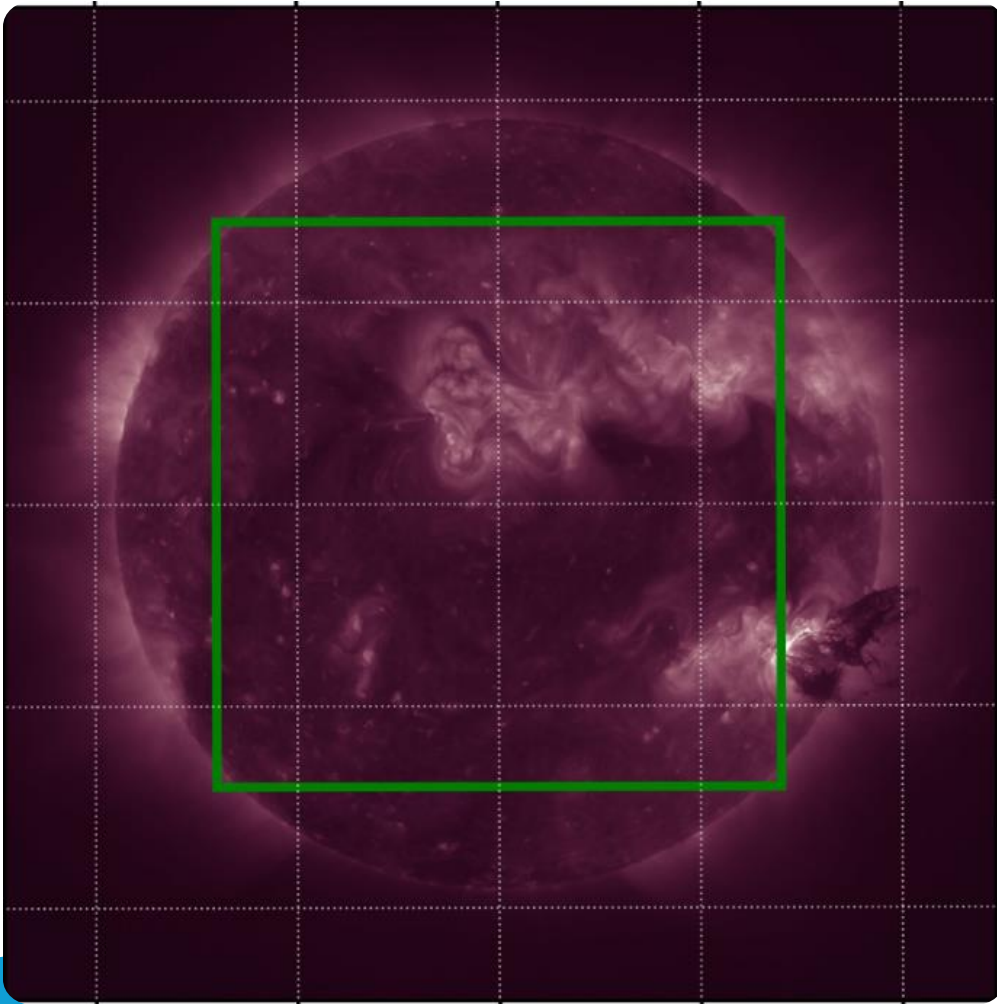
No cv. Training mapping does not translate to test mapping



Brown, E. J. E., Svoboda, F., Meredith, N. P., Lane, N., & Horne, R. B. (2022). Attention-based machine vision models and techniques for solar wind speed forecasting using solar EUV images. Space Weather

Forecasting 1 min average SWS at a 4 day lag from a single 211 Å image

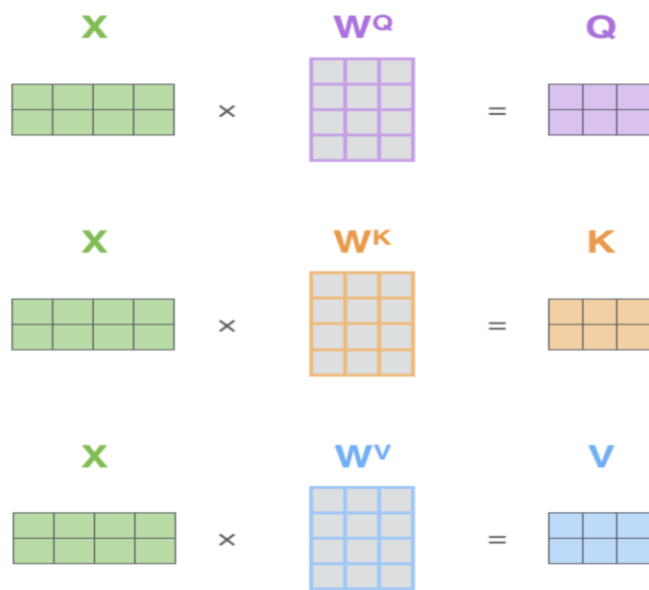




Input: A single image from the AIA 211A channel. Cropping the central 300^2 pixels from the SDOML dataset, and then rescale to 224 by 224.

Reasoning: Pixels outside this box are either background information or solar features at a faraway angle that would not impact the SWS of interest.

- BatchNorm: rescale the processed images because the pretrained feature extractors expect RGB images.\
- Attention mechanism:



Attention

$$Z = \text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

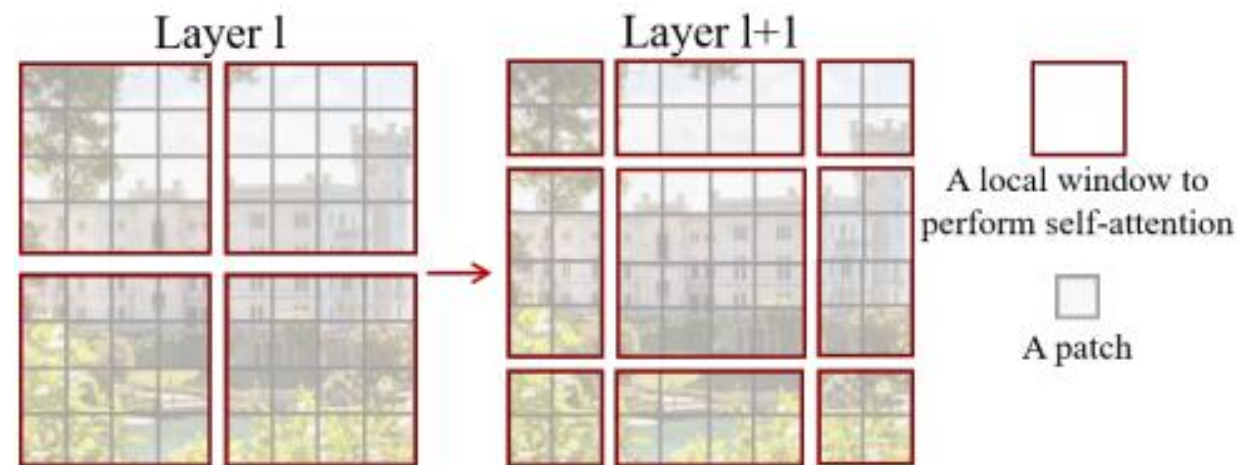
where Q, K, V is learnable parameters.¹³

The diagram shows the attention mechanism calculation:

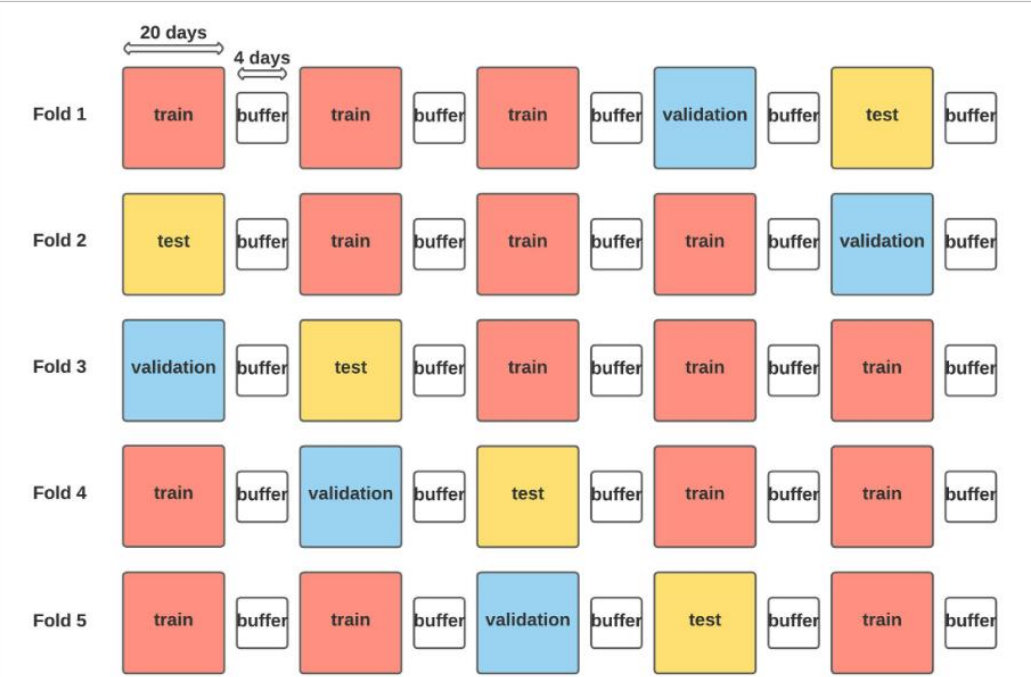
$$\text{softmax}\left(\frac{Q \times K^T}{\sqrt{d_k}}\right) \times V = Z$$

Where Q is a purple 2x4 matrix, K^T is an orange 4x2 matrix, V is a blue 2x4 matrix, and Z is the resulting pink 2x4 matrix.

- Vision Transformers (ViT): Embeddings come from images patches. For example, a $16 \times 16 \times 3$ pixel patch is firstly flattened to a 1×768 vector, and then multiplied by a trainable $768 \times n$ matrix to get a $n \times 1$ vector embedding (token). Combining all tokens into a single matrix X .
- Swin (Hierarchical Shifted Window) Transformer: Run attention in local windows to update pixel values, and shift window between layers so that one can get global interaction.



Training, validation, and testing

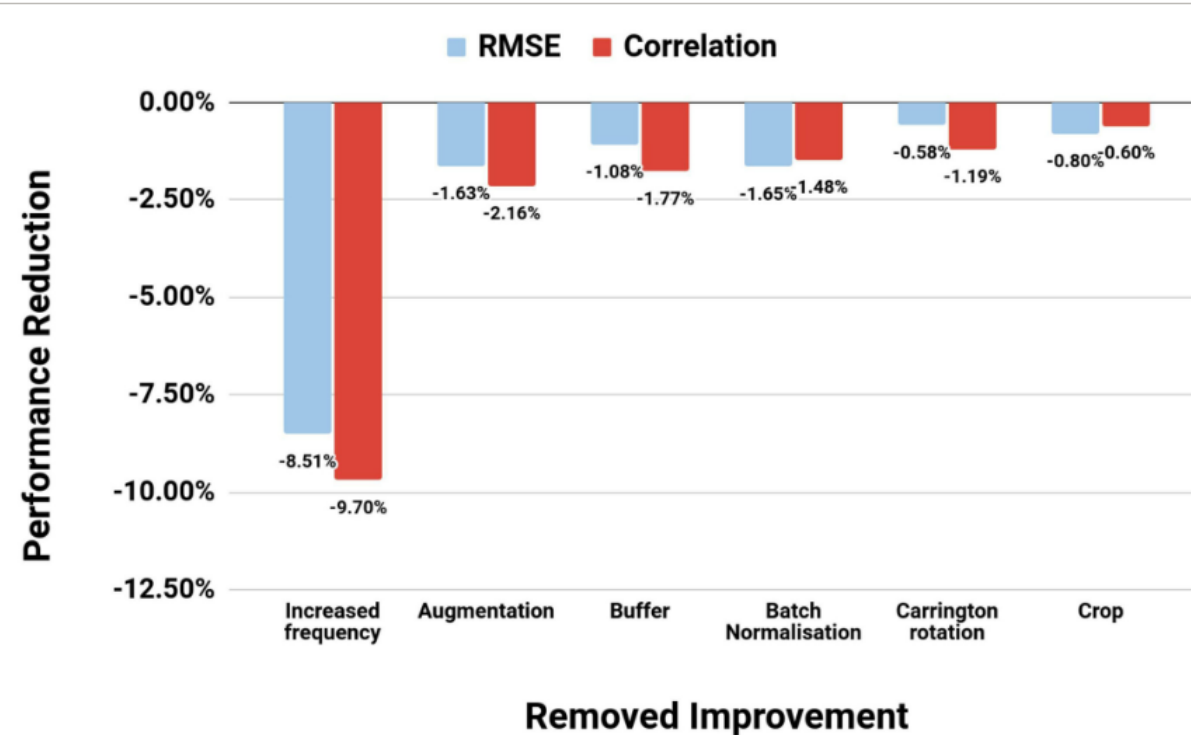


(a) 5-fold cross validation with buffer data thrown out. Pattern is repeated across the May 2010 to December 2018 range.

	RMSE	% Improvement	Correlation	% Improvement
Model				
Persistence (4 day)	127.59	-57.1%	0.080	-85.2%
Persistence (27 day)	100.86	-24.2%	0.426	-21.1%
Upendran et al. (2020)	81.21	-	0.54	-
Our models				
Solar InceptionNet v4	74.09	8.8%	0.609	12.7%
Solar DenseNet	73.92	9.0%	0.611	13.1%
Solar GoogleNet	73.71	9.2%	0.619	14.6%
Solar ResNet	73.52	9.5%	0.618	14.4%
Solar TNT	72.70	10.5%	0.629	16.5%
Solar Vision Transformer	72.66	10.5%	0.630	16.7%
Solar Swin Transformer	72.21	11.1%	0.634	17.4%

	RMSE	% Improvement	Correlation	% Improvement
Model				
Persistence (4 day)	118.76	-52.3%	-0.027	-104.9%
Persistence (27 day)	85.16	-9.2%	0.464	-15.6%
Raju and Das (2021)	78	-	0.55	-
Our model				
Solar Swin Transformer	71.65	8.3%	0.644	17.1%

Ablation study



(a) Performance reduction resulting from removing one improvement at a time

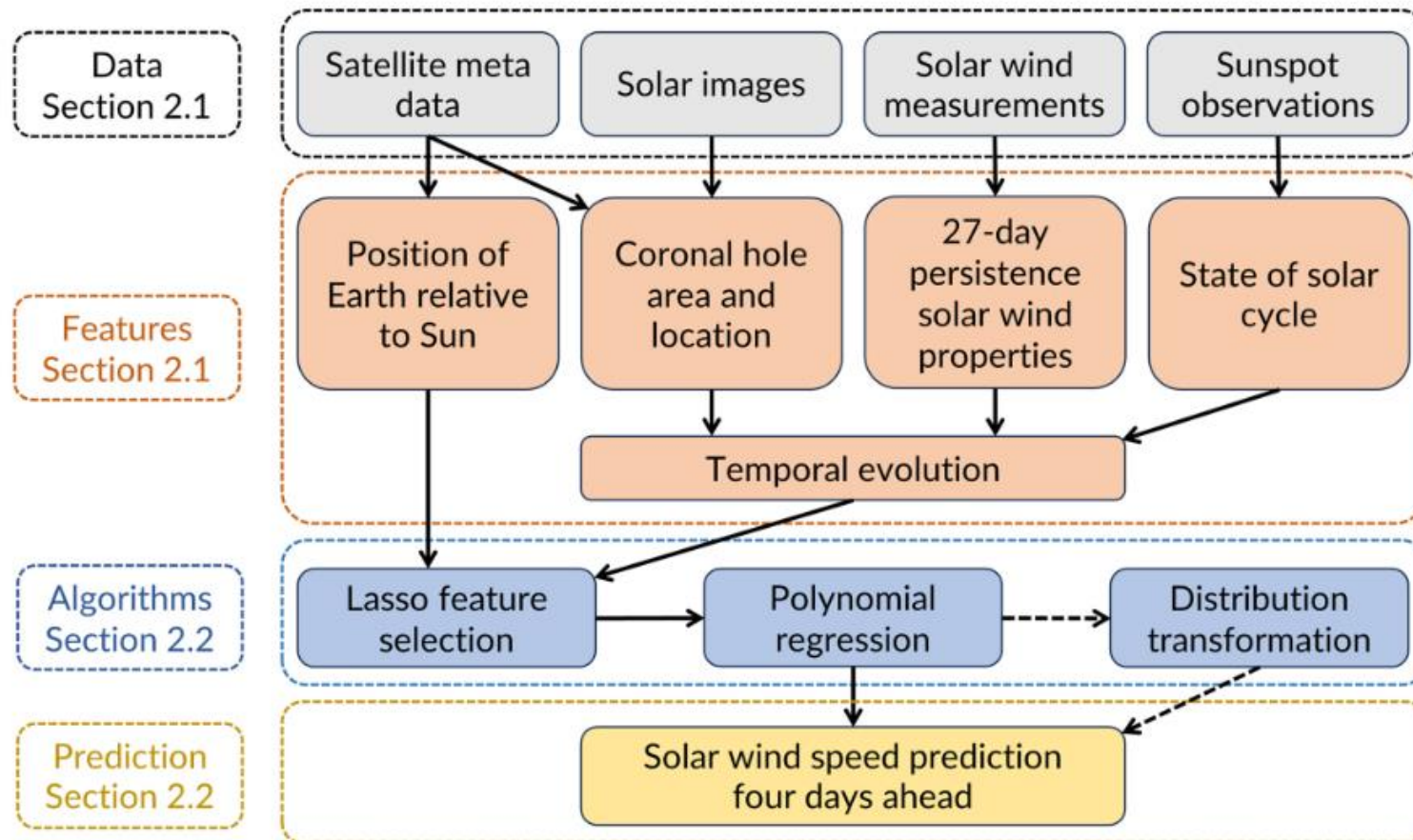
Frequency: cadence of prepared dataset, comparing 24hr to 30 min

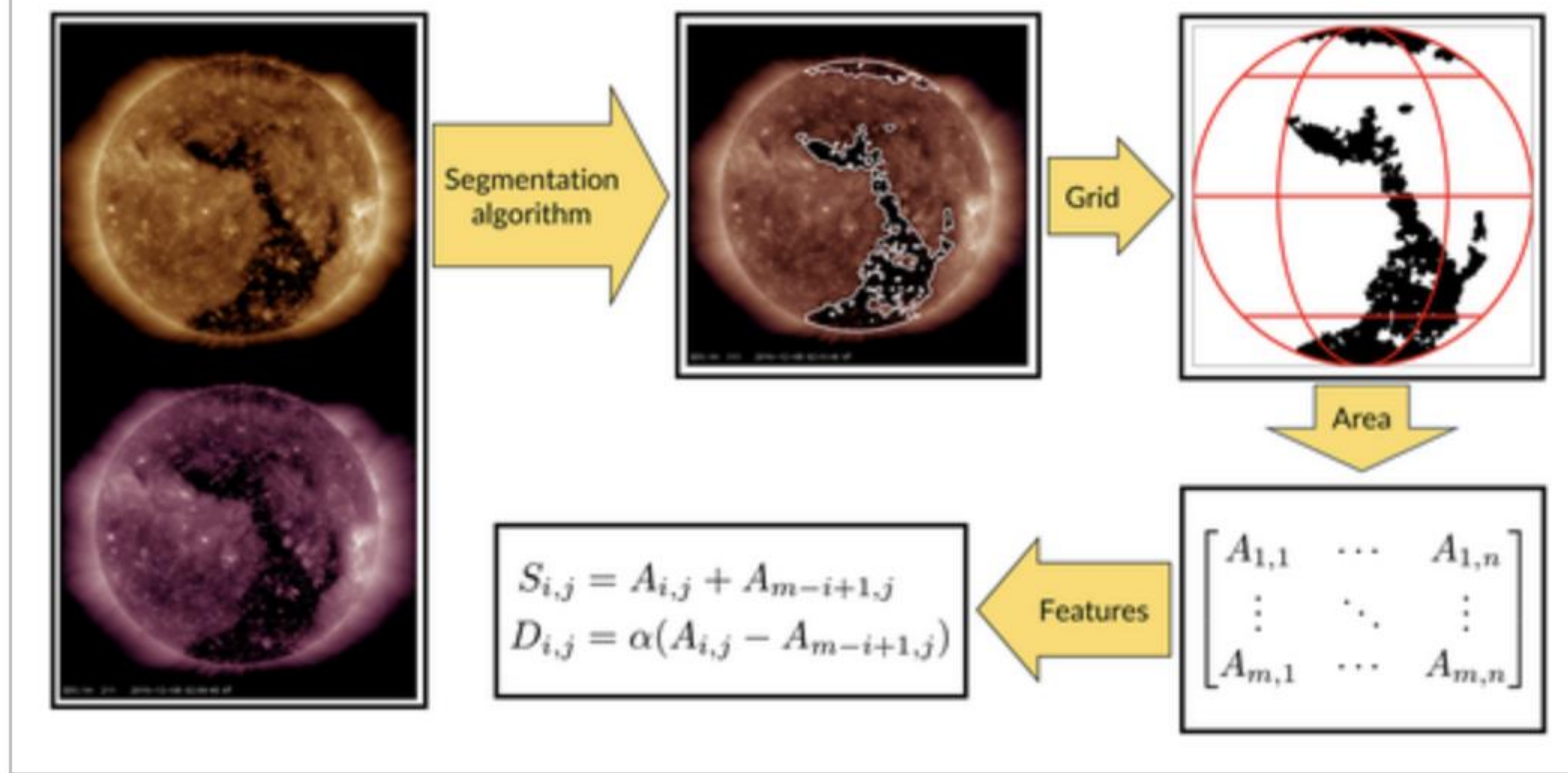
Augmentation: Flipping the training images north to south as extra data

Possible implementation

- Not open-source, but structure seems not overly complex, and datasets are publicly available. Training may take a long time (possibly hundreds of GPU hours)

Collin, D., Shprits, Y., Hofmeister, S. J., Bianco, S., & Gallego, G. (2025). Forecasting high-speed solar wind streams from solar images. Space Weather



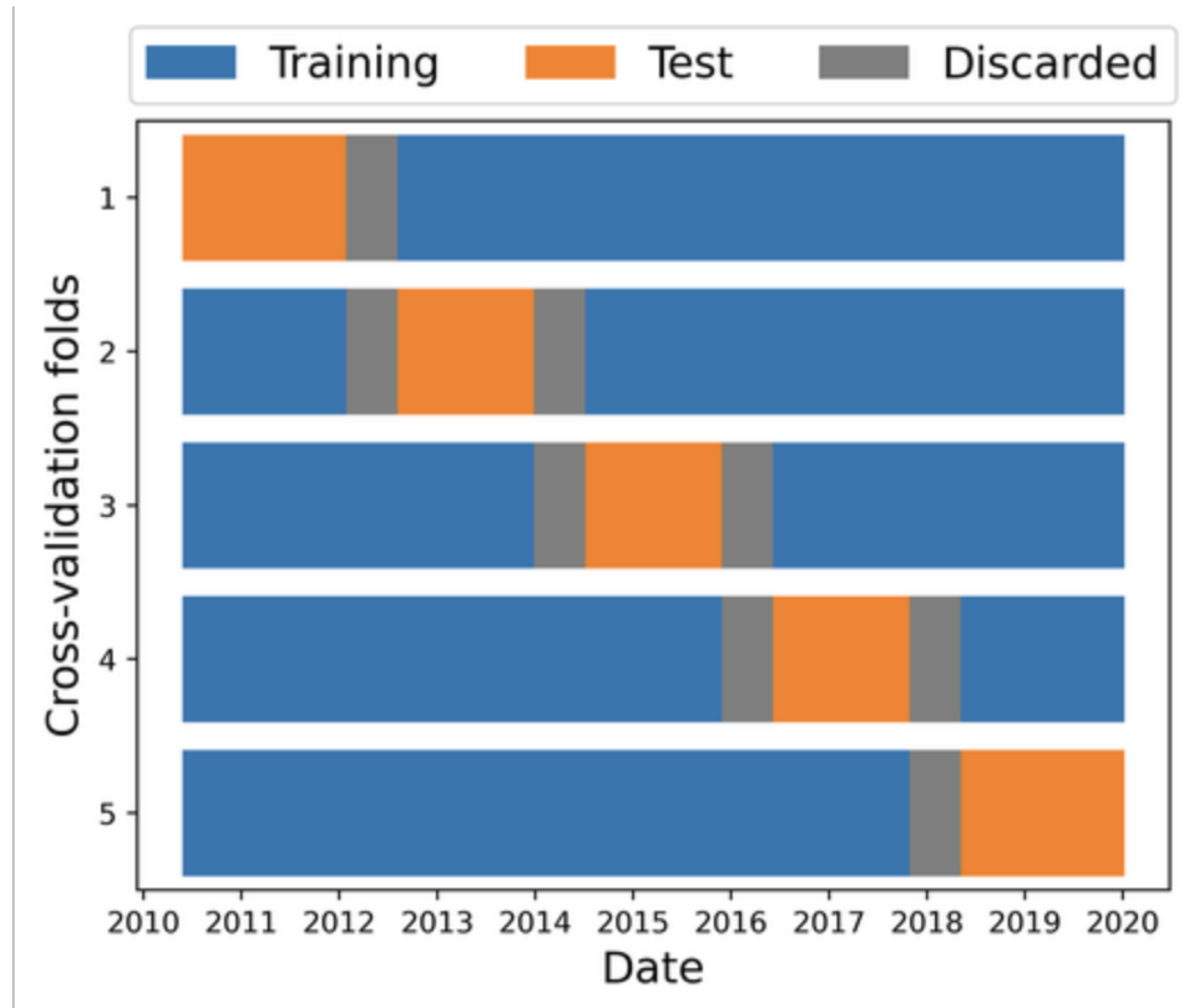


SWS from omniweb

Monthly sunspot number from the NOAA Space Weather Prediction Center

Spacecraft position in heliographic inertial coordinates of satellites at L1 from the OMNI_M data provided by the NASA COHOWeb.

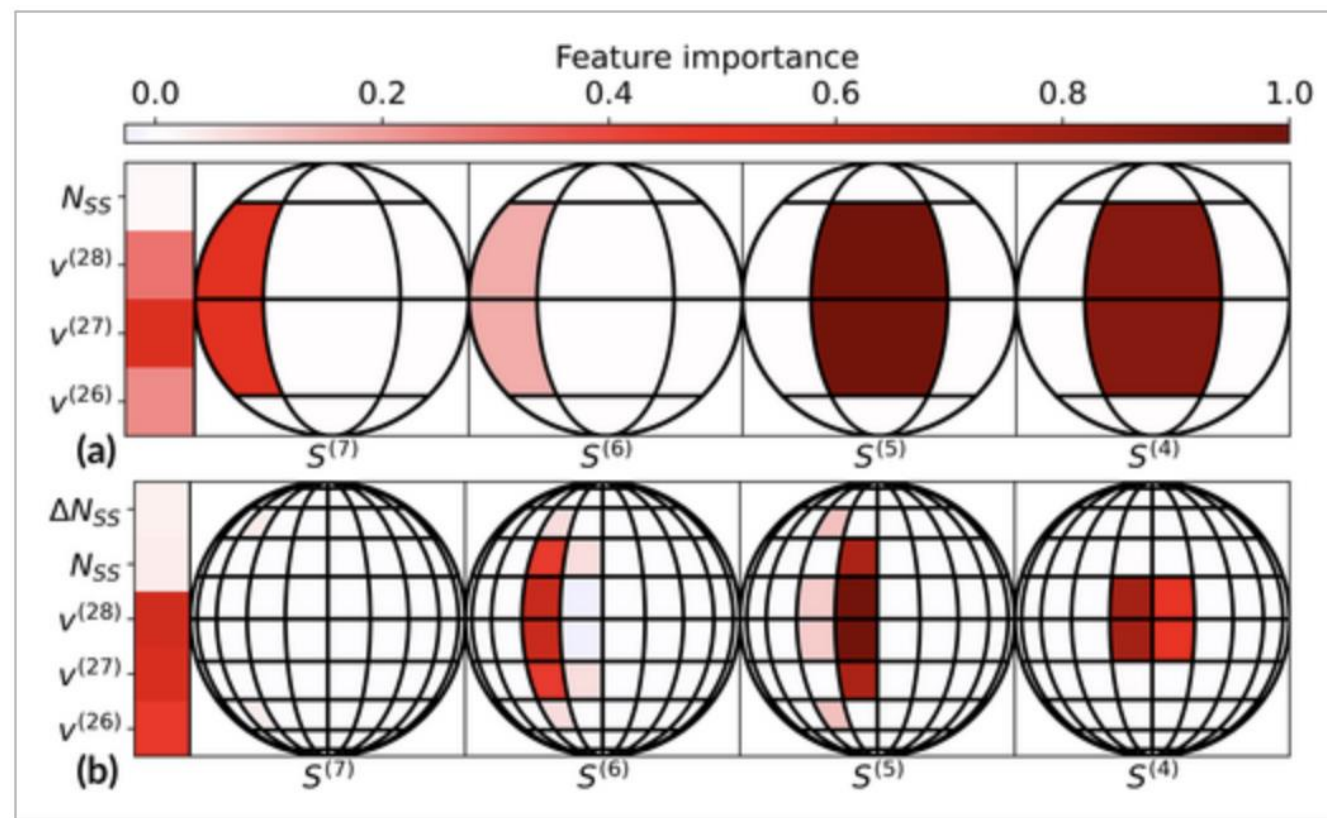
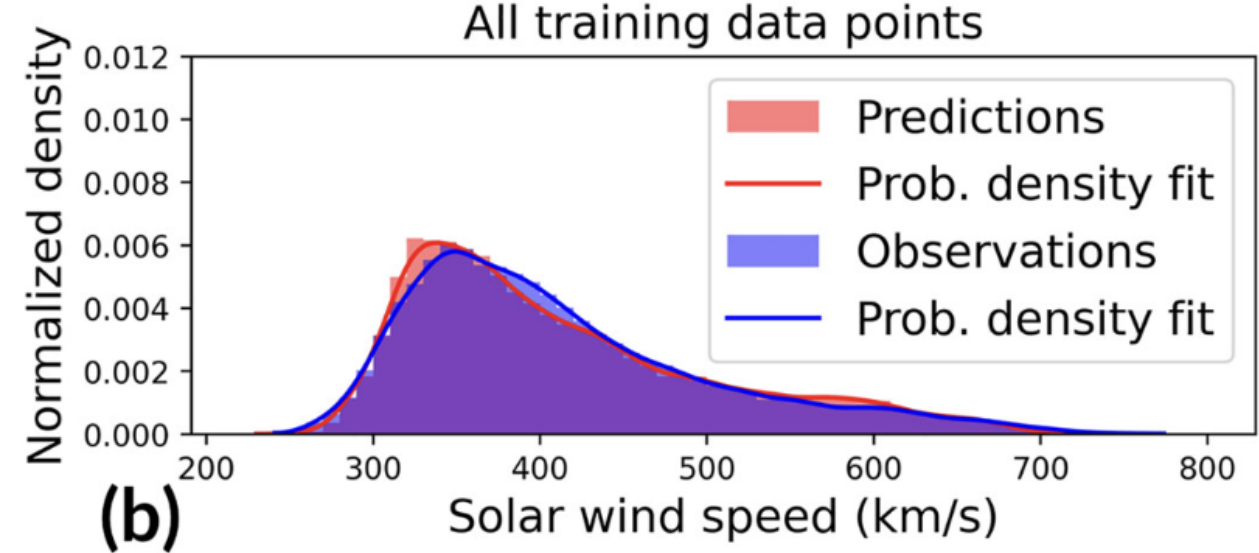
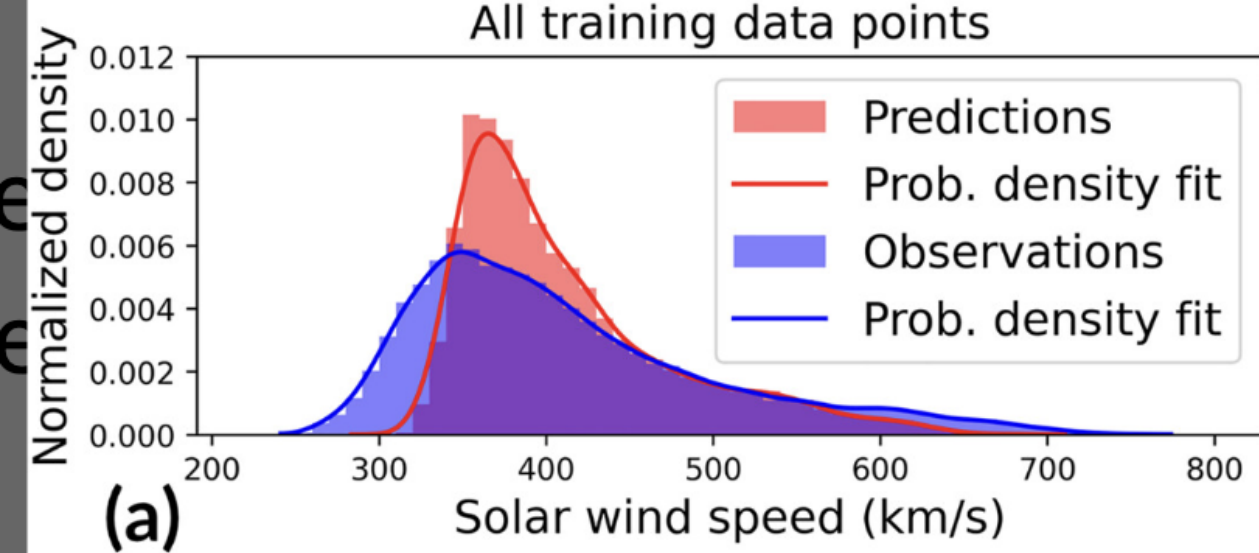
ICMEs affected periods removed. Cadence 1hr to predict 1 hr average SWS.



- applying a distribution transformation that maps the distribution of the output of the machine learning model onto the distribution of observations.
- Box-Cox transformation
- Computing lambda maximizing the probability of observing the given data under the assumption that the transformed data follows a normal distribution

$$y^{(\lambda)} = F_{\lambda}(y) = \begin{cases} \frac{y^{\lambda} - 1}{\lambda}, & \text{if } \lambda \neq 0 \\ \log(y), & \text{if } \lambda = 0 \end{cases}, \quad \lambda \in \mathbb{R},$$

$$\mathcal{A} \xrightarrow[\text{transform}]{\text{fit } \lambda_1} F_{\lambda_1}(\mathcal{A}) \xrightarrow{\text{normalize}} \tilde{F}_{\lambda_1}(\mathcal{A}) \approx \tilde{F}_{\lambda_2}(\mathcal{B}) \xrightarrow[\text{normalize}]{\text{inverse}} F_{\lambda_2}(\mathcal{B}) \xrightarrow[\text{transform with } \lambda_2]{\text{inverse}} \mathcal{B}$$



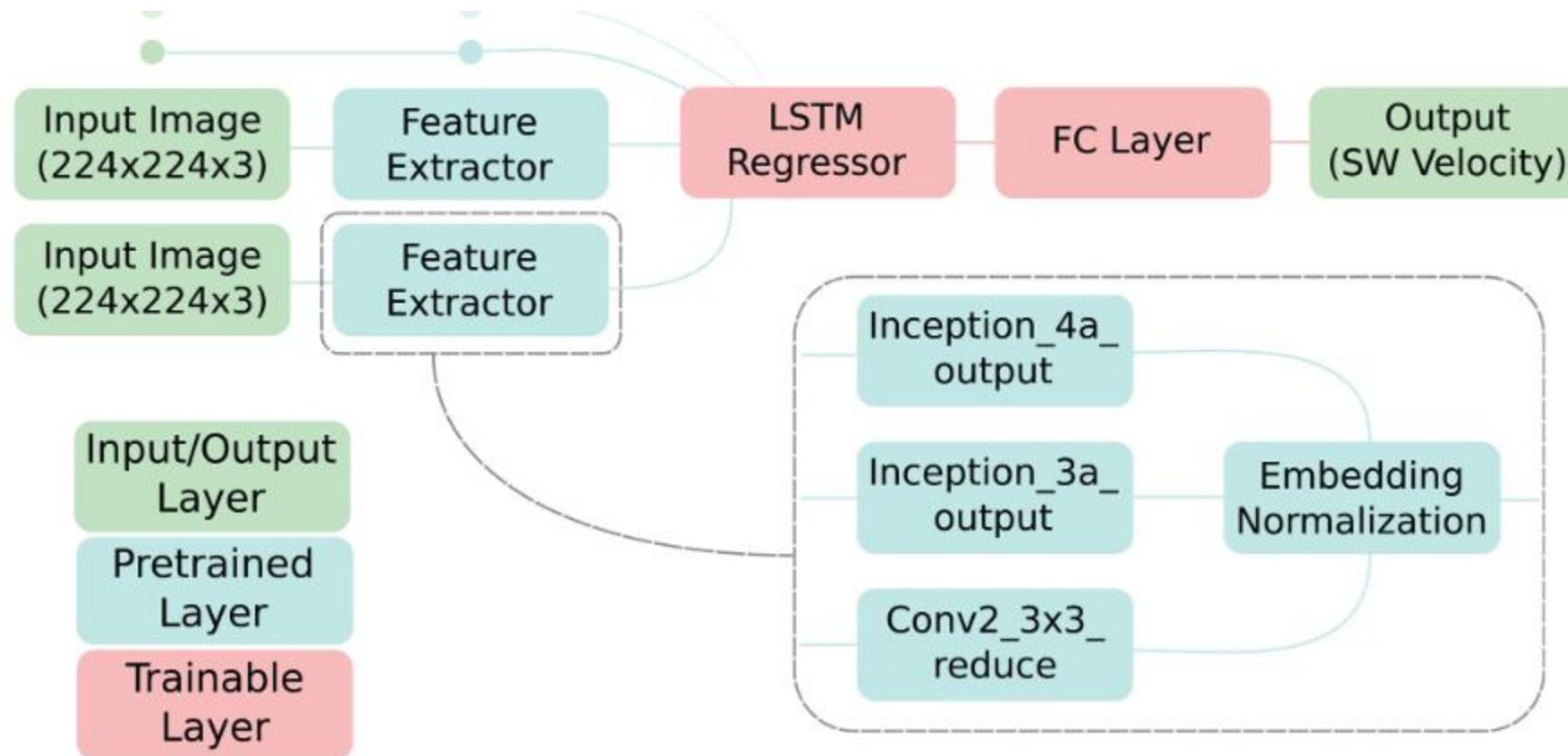
Model	Timeline			HSS peak velocities			HSS events			
	RMSE	MAE	CC	RMSE	MAE	CC	POD	FAR	TS	BS
AV	97.4	76.2	−0.30	–	–	–	–	–	–	–
27	89.1	65.4	0.56	78.4	59.7	0.54	0.71	0.26	0.57	0.95
CH	71.2	54.8	0.67	123.4	102.8	0.52	0.69	0.11	0.64	0.78
Ours	68.1	52.6	0.70	87.5	66.8	0.62	0.77	0.13	0.69	0.88
WSA	86.5	64.1	0.38	125.5	100.7	0.10	0.57	0.17	0.51	0.68
ESWF	91.8	69.5	0.43	104.6	81.6	0.23	0.68	0.35	0.50	1.05
Ours	61.8	47.4	0.55	121.4	101.4	0.34	0.68	0.12	0.63	0.77
EUHFORIA	90.8	68.8	0.36	105.1	87.4	0.28	0.64	0.30	0.50	0.91
Ours	62.9	48.0	0.68	68.6	53.8	0.71	0.88	0.12	0.78	1.00
GradBoost	75.6	59.1	0.65	106.6	84.1	0.60	0.67	0.16	0.60	0.80
Ours	71.2	55.3	0.70	95.4	73.2	0.59	0.74	0.14	0.66	0.86
SwinTrans	68.5	52.2	0.71	97.3	80.9	0.71	0.67	0.25	0.55	0.90
Ours	69.8	54.1	0.70	86.6	68.3	0.63	0.79	0.17	0.68	0.96

Possible implementation

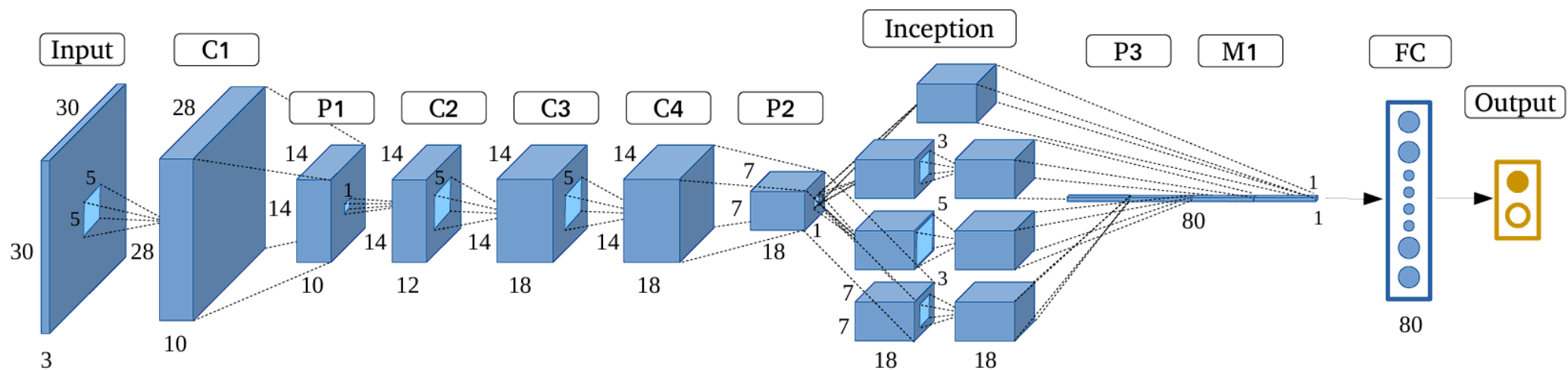
- Codes and dataset seems to be open source. And the model seems light-weight and fast to train.

Upendran, V., Cheung, M. C. M., Hanasoge, S., & Krishnamurthi, G. (2020). Solar wind prediction using deep learning. Space Weather

Sequence of SDO images at 193 A or 211A at 00:00, to predict daily averaged SWS days after.



- Input: From SDOML dataset, 512*512, resize to 224*224 and taking log for intensity, and replicate three copies to "simulate" RGB channels.
- Feature extractor: GoogleNet



Data from 1 January 2011 to 9 December 2018.

Feature extractor's weights are fixed, only train parameters of LSTM and one FC layer.

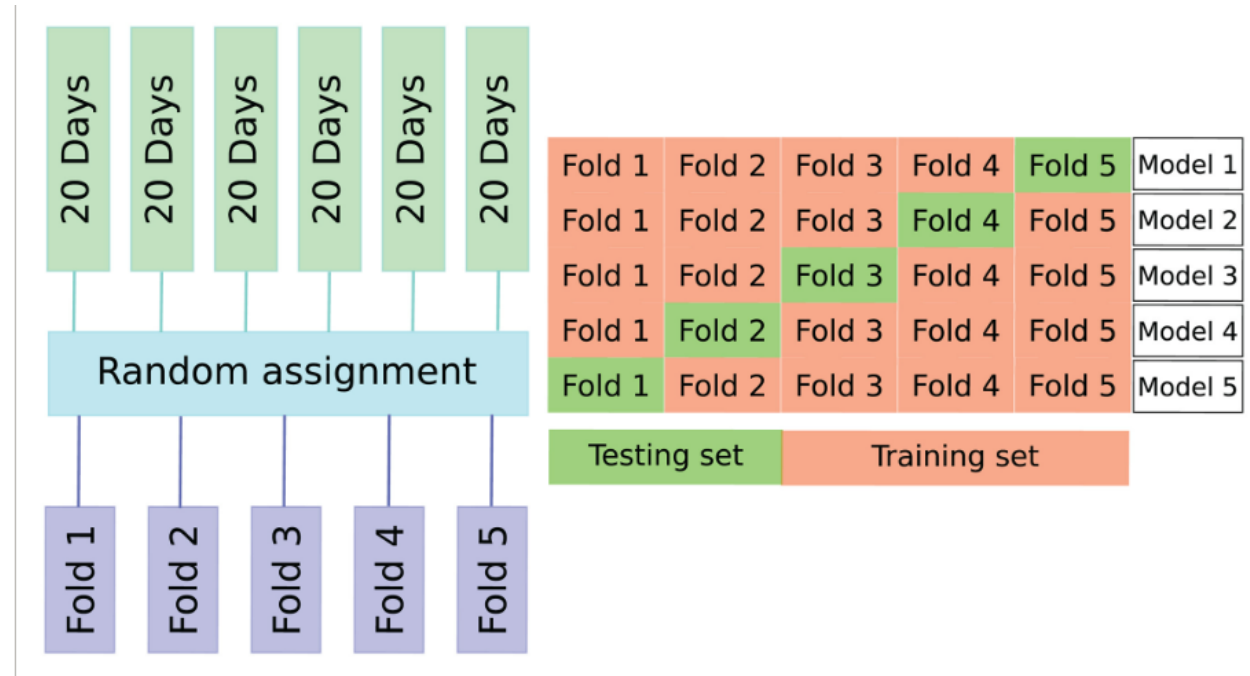


Figure 4

[Open in figure viewer](#)

[PowerPoint](#)

Training and test split for cross validation. First, the data are split into batches of 20 days each and each batch is randomly assigned to one of the cross-validation folds. Then, for one particular model (i.e., for one

Reports a correlation with observation of 0.61, RMSE of 76.4 km/s and χ^2 of 19.35, which I cannot find in the table they provided.

Table 5. RMSE Comparison of Our Model Predictions With the Benchmark Models

(H, D)	WindNet 193	WindNet 211	XGBoost	Persistence	SVM Linear	SVM RBF
4)						
(3, 1)	93.35 $\pm$ 4.33	82.60 $\pm$ 1.75	57.80 $\pm$ 0.80	113.81 $\pm$ 2.39	54.40 $\pm$ 1.01	54.34 $\pm$ 1.00
(3, 2)	88.17 $\pm$ 1.81	85.46 $\pm$ 2.63	84.10 $\pm$ 0.86	122.20 $\pm$ 3.13	78.59 $\pm$ 1.67	78.55 $\pm$ 1.63
(3, 3)	87.04 $\pm$ 1.25	83.97 $\pm$ 3.04	91.58 $\pm$ 1.05	125.16 $\pm$ 3.12	86.68 $\pm$ 1.76	86.75 $\pm$ 1.73
(3, 4)	87.21 $\pm$ 2.17	81.21 $\pm$ 1.86	92.72 $\pm$ 1.28	126.79 $\pm$ 2.92	88.72 $\pm$ 1.75	88.62 $\pm$ 1.80
(4, 1)	84.19 $\pm$ 2.83	80.27 $\pm$ 2.07	59.14 $\pm$ 0.82	122.20 $\pm$ 3.13	54.52 $\pm$ 1.03	54.48 $\pm$ 1.04
(4, 2)	86.42 $\pm$ 1.98	83.06 $\pm$ 2.51	83.78 $\pm$ 0.74	125.16 $\pm$ 3.12	78.47 $\pm$ 1.81	78.45 $\pm$ 1.78
(4, 3)	88.32 $\pm$ 1.93	80.28 $\pm$ 3.05	91.00 $\pm$ 1.25	126.79 $\pm$ 2.92	86.81 $\pm$ 1.59	86.82 $\pm$ 1.63
(4, 4)	82.93 $\pm$ 1.72	85.34 $\pm$ 3.10	92.34 $\pm$ 1.34	128.23 $\pm$ 2.96	88.87 $\pm$ 1.46	88.78 $\pm$ 1.58

Note. The 27 day persistence gives a RMSE of 93.14 $\pm$ 4.43.

Possible implementation:

- Seems open-source, data are all publicly available, and the architecture is not complex. But since the previous two works are more recent, reporting better performance metrics, and the metric from this paper does not match up, it probably makes more sense to choose from the previous two.